SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

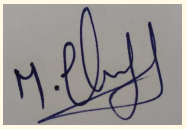
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	M.Virupaksha	**Reg. No.:**	192372205
Section / Batch:	CSE-AI(4th year)	**Date:**	06/10/2026

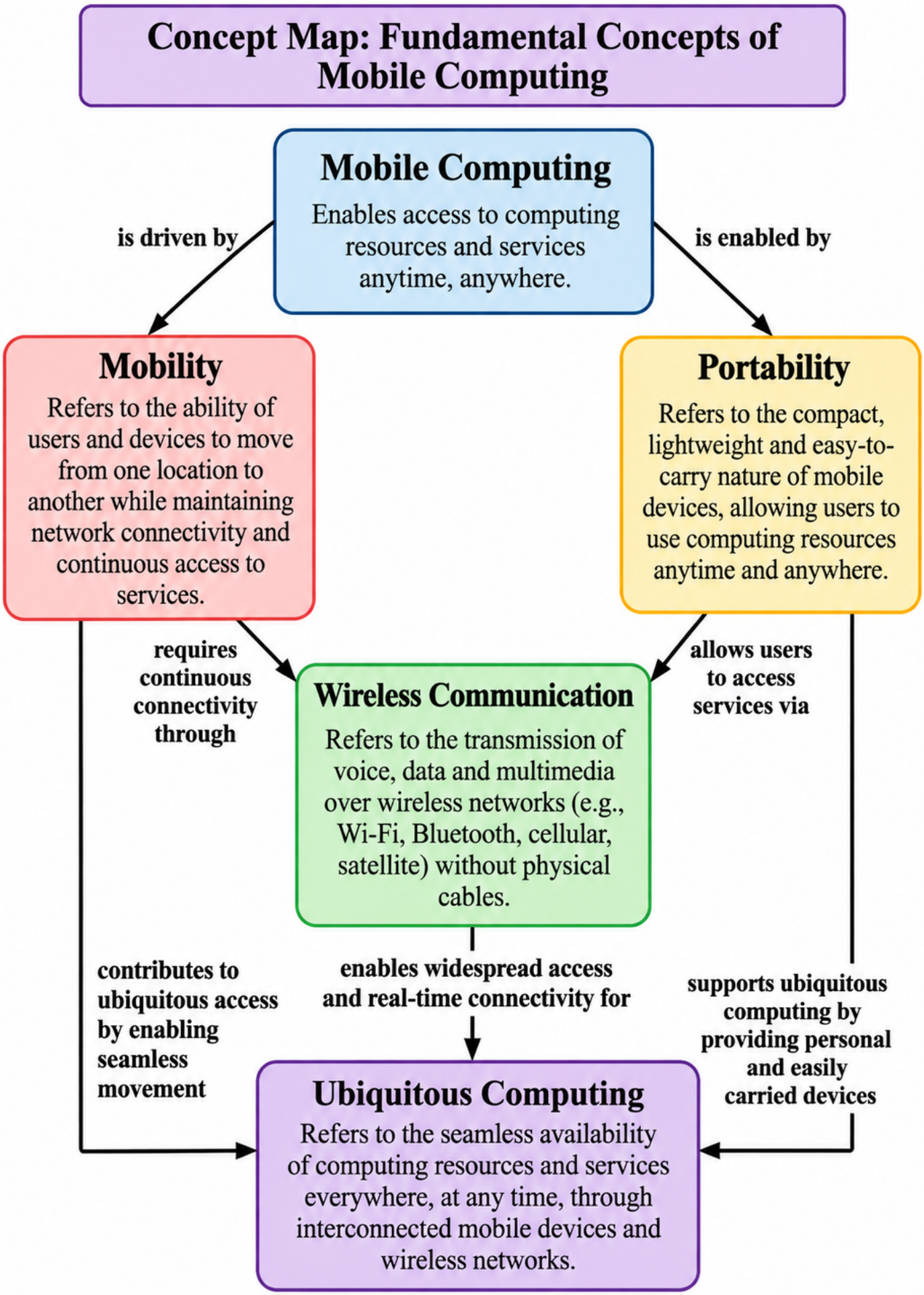
Code of Conduct

I M.Virupaksha(192372205) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)


Concept Map: Fundamental Concepts of Mobile Computing
Mobile Computing
Enables access to computing resources and services anytime, anywhere.
is driven by
is enabled by
Mobility
Refers to the ability of users and devices to move from one location to another while maintaining network connectivity and continuous access to services.
Portability
Refers to the compact, lightweight and easy-to-carry nature of mobile devices, allowing users to use computing resources anytime and anywhere.
requires continuous connectivity through
allows users to access services via
Wireless Communication
Refers to the transmission of voice, data and multimedia over wireless networks (e.g., Wi-Fi, Bluetooth, cellular, satellite) without physical cables.
contributes to ubiquitous access by enabling seamless movement
enables widespread access and real-time connectivity for
supports ubiquitous computing by providing personal and easily carried devices
Ubiquitous Computing
Refers to the seamless availability of computing resources and services everywhere, at any time, through interconnected mobile devices and wireless networks.

EVALUATION TABLES – FUNDAMENTAL CONCEPTS OF MOBILE COMPUTING

1. Priority of the Concepts

The ranking reflects each concept's direct contribution to enabling mobile computing. The concepts depend on one another; the order indicates functional importance, not that one concept can operate entirely alone.

Priority	Concept	Contribution and significance
1	**Wireless communication**	Provides the data link between a mobile device and network services without a physical cable. Without wireless connectivity, remote communication while moving would be severely limited.
2	**Mobility**	Allows users or devices to change location while maintaining access to communication and services. It supports continuity and roaming.
3	**Portability**	Makes computing practical on the move because devices can be carried and used in different places. Portability alone does not guarantee network connectivity.
4	**Ubiquitous computing**	Represents the broader goal of computing being available across everyday environments. It depends on connectivity, usable devices and helpful services.

2. Justification of Linking Phrases

Linking phrase	Justification
“Wireless communication enables mobile computing”	Wireless networks carry data between devices and network resources, allowing communication without a wired connection.
“Mobility relies on wireless connectivity”	When users change location, wireless access and handover help maintain communication where coverage is available.
“Portability makes mobile access practical”	Compact, battery-powered devices can be carried and used in different places, making on-the-go computing convenient.
“Ubiquitous computing extends access”	Connectivity, portable devices and useful services together make computing accessible across everyday situations.
“Supporting factors influence reliability”	Battery capacity, network coverage, software and security affect availability, performance and user trust.
“Together improve user experience”	The concepts complement one another to provide flexible access, continuity, convenience and productivity.

3. Overall Evaluation

Wireless communication is ranked first because it supplies the connection needed for remote access. Mobility and portability make that connection useful across different locations. Ubiquitous computing is the broader outcome: services become more continuously available when the other concepts and supporting technologies work together. Effective mobile computing depends on the combination, not on one concept in isolation.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____